

BSCI 502, Introduction to Biotechnology Manufacturing
Fall 2018
Online/Accelerated

Course Description

Introduction to the terminology and practices of the biotechnology industry, with an emphasis on the manufacturing, regulatory, and legal aspects of modern biotechnology.

Instructors

Primary	Dr. Becky Robinson-Zeigler	rebecca.robinson-zeigler@unh.edu
Backup	Dr. Richard McFarland	richard.mcfarland@unh.edu

Office Hours

By appointment and virtual (i.e., via telephone or video conference)

Objectives

- To gain an understanding of the types of modern biotechnology products and how they are manufactured
- To develop a basic knowledge of the science underpinning biotechnology products
- To gain an understanding of basic manufacturing principles
- To develop a basic knowledge of the product development process
- To understand the regulatory framework for determining the safety and effectiveness of biotechnology products

Textbook

None. Readings, videos, and other content will be assigned in each online module.

Course Structure

Each week, one module will be activated and you will have approximately one week to complete the assignments contained in that module. The module will include videos or lectures to watch and articles to read. There are seven (7) course modules; when the seventh module is complete, the course is over.

Assignments

Weekly Discussion Posts

Each module includes one or more discussion topics that will reinforce the presented content. You are expected to fully participate by posting a response to the discussion topic(s). Your post will demonstrate your ability to apply what you have learned in the module by crafting a response rooted in your understanding of the module material. Responses need to be your original work supported by the module materials and other outside supportive sources. You are also expected to respond each week to at least **two** of your classmates' postings with value-added comments and/or follow-up questions.

Given the online nature of this course, the Academic Honesty Policy (below) is critical. Please do not copy/paste answers from the internet or other sources without proper attribution.

The initial post (i.e., your response to the discussion topic) is due **Wednesday, 11:59 PM ET**. The responses to your classmates' postings are due **Friday, 11:59 PM ET**. There are two exceptions to these due dates explained below.

***Module 1 Discussion Posts:** This module has no classmate response requirement, therefore, the posts for this module are due **Friday, 11:59 PM ET**.

****Module 7 Discussion Posts:** Due to the extra content in this module, the initial posts will be due **Thursday, 11:59 PM ET**. The responses to your classmates are still due **Friday, 11:59 PM ET**.

*****Late Penalty:** Any discussion posts or responses submitted after the due date, but within a 24-hour period will receive a 1-point reduction. Any discussion posts or responses submitted one day after the due date will receive a 2-point reduction. Any discussion posts or responses submitted two days after the due will receive a 4-point reduction. Any discussion posts or responses submitted three days after the due date will receive a **zero**.

Quizzes

The course includes eight (8) graded, timed, “open-book” quizzes to assess your learning progress in the course. Modules 1-6 include one quiz and Module 7 includes two quizzes to account for the additional content included in this module. Each quiz consists of multiple-choice, true/false, matching, or fill-in-the-blank questions based on the content in the module. You will have one hour to complete and submit the quiz. Students will be able to view the correct answers immediately after the due date.

Your work on the quizzes should be your own. Do not discuss your answers with other students before submitting your responses, or before they have submitted theirs.

Quizzes are due **Friday, 11:59 PM ET**.

*****Late Penalty:** Any quiz submitted one day after the due date will receive a 5-point reduction. Any quiz submitted two days after the due will receive a 10-point reduction. Any quiz submitted three days after the due date will receive a **zero**.

Grading

Your course grade will be determined based on the percentage of points earned out of the possible 480.

Course Grade Breakdown (subject to additions/changes)

Breakdown by Module		Breakdown by Tasks	
6 modules @ 50 points each	300	Initial and Response Posts	154
1 module @ 100 points	100	Quizzes	246
Participation	80	Participation	80
Total Points Available	480	Total Points Available	480

Grade Scale Based on Percentage of Points Earned Out of the Possible 480 Points

A	94 – 100 %	C	72 – 76 %
A-	90 – 93 %	C-	70 – 72 %
B+	87 – 89 %	D+	67 – 69 %
B	83 – 86 %	D	63 – 66 %
B-	80 – 82 %	D-	60 – 62 %
C+	77 – 79 %	F	59 and below

Grades are **NOT** rounded. For example, 89.9% is a B+.

Extra Credit: No extra credit is offered.

Weekly Discussion Posts

Discussion posts for Modules 2-7 will be worth a total of 20 points. You can receive a total of 12 points for the initial discussion post and 8 points for the two responses to your classmates' posts. The rubric for grading of the discussion posts is stated below:

Initial Post	
6 points	Posted a response to the discussion topic
2 points	Posted response addresses most of the discussion topic and includes a reference (if applicable)
2 points	The included reference is from a primary source (i.e., a published journal article, standard, guidance document, or guideline) (if applicable)
2 points	The posted response provides insightful commentary that demonstrates an understanding of the topic and the module content
12 points	Total points possible

Response Posts	
2 points	Posted response to one classmates' post
2 points	Posted response to a second classmate's post
4 points	Posted response(s) to classmates provide insightful commentary that supports your position of agreement or disagreement. Responses go beyond surface level and superfluous comments that comment on other student's effort rather than the discussion points they are making. Responses should NOT look like: "nice job" or "I agree". Responses should look like deep thinking such as "I agree/disagree because..." (supported by evidence from the content presented) or "I wonder if..." (applying an idea to a new application or scenario).
8 points	Total possible points

Discussion posts for Module 1 will be worth 14 total points. Points for this post will be awarded based on effort only; if you complete the posts for Module 1, you will receive all 14 points.

Participation

Participation points are earned based on your activity in the discussions for each module. Participation in the discussions is extremely important. First, it gives the instructor an indication of your appreciation of, understanding of, and ability to apply the concepts presented in the modules. Second, it enables each of you to interact with your classmates in a knowledge-building arena similar to the structure that is currently taking place in the nascent tissue engineering industry. The ability to challenge, form, and defend your own conceptual claims among peers through collaboration and consensus of opinions based on the current evidence is integral to your success in the fast-paced and constructive real-world R&D environments.

You can earn up to 10 participation points for each discussion topic.

- If you complete each portion of the assignment—initial post and two responses—you will receive **10 points**.
- If you complete only a portion of the assignment, you will receive **5 points**.
- If you do not complete the assignment, you will receive **zero** participation points for that discussion topic.

Quizzes

Grades for quizzes are based on your correct answer to the multiple choice, true/false, matching, or fill-in-the-blank questions. Quizzes have anywhere from 12-18 questions. Questions vary from 1-6 points each. The quiz will display the point value for each question. All quizzes for Modules 2-7 are worth a possible 30 points. The quiz for Module 1 is worth a possible 36 points.

Disabilities

If you have a documented disability or believe you may have a disability that requires accommodations, please contact Disability Services Coordinator, Janessa Zurek (603-641-4383, janessa.zurek@unh.edu) to coordinate a reasonable accommodation.

Academic Support/Tutoring

The Center for Academic Enrichment (CAE) [2nd floor, Learning Commons] is a resource for all students enrolled in UNH Manchester courses. Their mission is to help under-prepared students prepare, prepared students advance, and advanced students excel. Trained peer tutors and professional staff provide free assistance with reading and writing in all courses, as well as content in all math and science courses. Support for other content areas is dependent on the background and competencies of current peer tutor staff. All enrolled students are entitled to one (1) hour of free one-to-one tutoring per week, per course. In addition, peer and professional tutors may run study groups for your class and provide drop-in math tutoring. Please visit the CAE online at <http://manchester.unh.edu/current-students/cae-more-powerful-learning> to learn more about their services or to make an appointment with a tutor.

MyCourses (Canvas) and E-mail

Check MyCourses every day. Non-urgent announcements will be posted on MyCourses. Urgent/time-sensitive announcements will be sent, when possible, via e-mail. Check your UNH e-mail account often as this is the one that will be used to contact you outside of class.

Academic Honesty

The Biology and Biotechnology Programs at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<https://www.unh.edu/student-life/09-academic-honesty-2017>). By submitting any piece of work in this course, you declare that you have read and understand the policy, and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work—in whole or in part—as your own; collaborating on answers for individual assignments, or allowing your own work to be used by another student; copying information from a website or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in lab, bringing crib notes into an exam or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam. All electronic devices must be turned off and put away for the duration of any quiz or exam. Ear buds are not permitted.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Class & Assignment Schedule

Module	Assignment	Due Date
Module 1 – Introduction to Manufacturing and Biotechnology Medical Products	Discussion post – “Reflection”	Friday, 08/31, 11:59 PM ET
	Discussion post – “Introductions”	Friday, 08/31, 11:59 PM ET
	Quiz 1 – “Manufacturing, Scope, History of Biotech”	Friday, 08/31, 11:59 PM ET
Module 2 – Manufacture of Monoclonal Antibodies and Therapeutic Proteins	Initial discussion post – “Bioavailability of mAbs and therapeutic proteins”	Wednesday, 09/05, 11:59 PM ET
	Response posts	Friday, 09/07, 11:59 PM ET
	Quiz 2 – “Immunology, Cell Culture, Manufacture of mAbs and Proteins”	Friday, 09/07, 11:59 PM ET
Module 3 – Manufacture of Cell Therapies	Initial discussion post – “The major challenges to large-scale manufacturing of cells and ways to overcome them”	Wednesday, 09/12, 11:59 PM ET
	Response posts	Friday, 09/14, 11:59 PM ET
	Quiz 3 – “Cell Bio, Stem Cells, Manufacture of Cell Therapies”	Friday, 09/14, 11:59 PM ET
Module 4 – Manufacture of Gene Therapies	Initial discussion post – “The risks and challenges of gene therapy products”	Wednesday, 09/19, 11:59 PM ET
	Response posts	Friday, 09/21, 11:59 PM ET
	Quiz 4 – “Genetics, Vectors, Manufacture of Gene Therapies”	Friday, 09/21, 11:59 PM ET
Module 5 – Manufacture of Regenerative Medicine Products	Initial discussion post – “The challenges to advancement of the field of regenerative medicine”	Wednesday, 09/26, 11:59 PM ET
	Response posts	Friday, 09/28, 11:59 PM ET
	Quiz 5 – “Tissue Engineering, Regenerative Medicine, Biomaterials, Bioprinting”	Friday, 09/28, 11:59 PM ET
Module 6 – Current Good Manufacturing Practices (cGMPs) and Standards	Initial discussion post – “The role of standards in cGMPs and SOPs”	Wednesday, 10/03, 11:59 PM ET
	Response posts	Friday, 10/05, 11:59 PM ET
	Quiz 6 – “cGMPs, Standards”	Friday, 10/05, 11:59 PM ET
Module 7 – Biotechnology Product Development: From R&D to the Market	Initial discussion post – “Biotechnology products and intellectual property”	Thursday, 10/11, 11:59 PM ET
	Initial discussion post – “The biggest hurdle in getting biotechnology products to market”	Thursday, 10/11, 11:59 PM ET
	Response posts	Friday, 10/12, 11:59 PM ET
	Quiz 7 – “Preclinical, Clinical, IP”	Friday, 10/12, 11:59 PM ET
	Quiz 8 – “FDA, TPLC, Combo Products, Marketing Applications, Reimbursement”	Friday, 10/12, 11:59 PM ET